BHARGAVI SINGH

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B.Tech CS (Cloud Computing) student at SRM with a strong foundation in AI/ML, data engineering, and cloud infrastructure. As Corporate Lead for SRM's Mars Rover Team, I managed external PR and social media presence, and authored official competition documentation, including Preliminary Design Reports for international events. I am able to explain complex problems to both technical and non-technical stakeholders. AWS and IBM certified with hands-on experience in data analysis and visualization. Placed 4th globally at the International Rover Challenge 2026.

Education

B.Tech Computer Science & Engineering (Cloud Computing)
SRMIST, KTR
8.9/10 as of August 2026

Skills

Leadership| Team Management | Public Speaking | Marketing | Project Management | JAVA | Communication | Designing| Problem Solving| AWS | Data Visualization | Deep Learning| Data Analytics | DBMS | AI / ML | Java | HTML/CSS | SQL | Python | C | Cyber Security | Team Work | Linux | software development cycle | Web Development | UI/UX | Canva | Figma

Recognitions

- NPTEL Star April 2026
- International Rover Challenge 2026 — 4th Place Globally + Winner, Astrobiology Expedition
- Techknow 2023 — Automatic Solar Tracker
- Best Paper -ICCPCT 26
- Young Journalist (YJ) — Reporting, anchoring, and news editing for Ryan TV

Experiences

- Freelance Brand Designing, Logo and card designing
- Cloud Virtual Intern | Amazon Web Services | 2024
- Cyber Security Virtual Intern | Palo Alto Networks | 2024
- Corporate Lead | Team RUDRA SRM Mars Rover Team | 2025 – Present
- Health Outreach Volunteer | Pinkish — Women's Health & Hygiene Organisation | 2022–2023

Languages

English | Hindi | Japanese

Publications

- Chapter on Data structures and Algorithm applications in Contemporary Research in Engineering, Management and Science
- IEEE: PsyPredict - An EVM based Postpartum Depression Predictor
- IEEE: MOODFIT: GPU Accelerated Intelligent Outfit Recommendation System

Projects

Postpartum Depression Predictor — Python · XGBoost · SVM · Neural Nets · Hadoop · HTML

▸Pioneered an ML pipeline analyzing 1,500+ postnatal records to detect postpartum depression risk early, which is critical in India, where 22% of mothers are affected yet often undiagnosed.▸ Overcame class imbalance via advanced preprocessing; benchmarked XGBoost (92% accuracy), SVM (88%), neural networks, and ensemble voting (94% accuracy) for superior prediction. ▸ Deployed a real-time interactive dashboard empowering stakeholders to monitor, visualize, and act on insights efficiently. https://github.com/bgvis9125/PPD

Real-Time Sign Language Translator — Python · OpenCV · TensorFlow · Computer Vision

▸ Designed and trained an object detection model on a self-curated hand-gesture dataset captured via webcam with no pre-labeled data required, addressing communication barriers for the 466M+ deaf and hard-of-hearing people worldwide who face 80% unemployment rates due to inaccessible communication.▸Achieved real-time inference using TensorFlow Object Detection API, making assistive communication accessible without specialist hardware. https://github.com/bgvis9125/Sign-Language-Translator

AI Outfit Recommendation Assistant — Python · HTML · CSS · GPU · API Integration

▸ Built a context-aware styling chatbot that factors in mood, weather, and occasion to generate personalised recommendations, helping users express themselves better on tough days.▸Implemented persistent memory across sessions so the assistant learns user preferences over time, increasing relevance without retraining. https://github.com/bgvis9125/MoodFit

Airport Path Optimizer — HTML · CSS · Graph Algorithms

▸Implemented Prim's Algorithm to compute Minimum Spanning Trees for airport flight path scheduling with live visual path rendering, tackling the $800B aviation industry challenge where delays/fuel waste affects 5-10% of flights daily [IATA Economics]. ▸ Reduced route complexity to a data-structure problem, demonstrating ability to translate algorithmic theory into practical optimization tools for aviation efficiency. https://github.com/bgvis9125/Airport-Path-Optimizer.

Sample Analysis on the Mars rover - Computer Vision · Astrobiology

▸Contributed to the development of rover-based astrobiology systems for the International Rover Challenge (IRC) 2026, enabling collection, analysis, and characterization of environmental samples to assess microbial habitability and identify potential biological indicators.▸ Supported the integration and validation of onboard scientific workflows where the rover functioned as a mobile laboratory, performing in-situ sample investigations under Mars-analog mission conditions.

Certifications

- AWS Certified: AI Practitioner, Cloud Practitioner, Data Engineer, Solution Architect
- IBM: Python for Data Science, Cloud Essentials, Containers, Multicloud
- NPTEL: Database Management, Cloud Computing, Networking, Applied Accelerated AI, High-Performance Scientific Computing